

Ultrafast Laser-Induced Spin Dynamics in All-Semiconductor Ferromagnetic CrSBr–Phosphorene Heterostructures

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Cite This: *J. Phys. Chem. Lett.* 2025, 16, 782–788



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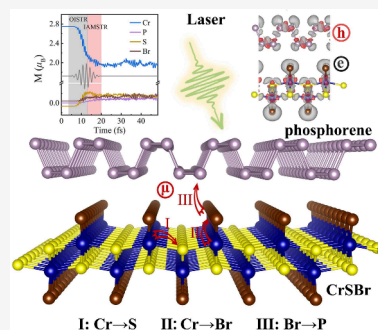


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Supporting Information

ABSTRACT: Ultrashort laser pulses are extensively used for efficient manipulation of interfacial spin injection in two-dimensional van der Waals (vdW) heterostructures. However, physical processes accompanying the photoinduced spin transfer dynamics on the all-semiconductor ferromagnetic vdW heterostructure remain largely unexplored. Here, we present a computational investigation of the femtosecond laser pulse induced purely electron-mediated spin transfer dynamics at a time scale of less than 50 fs in a vdW heterostructure. The latter is composed of two semiconducting monolayers, namely, a ferromagnetic material CrSBr and a nonmagnetic phosphorene, and is denoted as CrSBr-P. We observe an ultrafast spin injection from the Cr atoms to the P atoms in a few femtoseconds by both optically induced and interfacial atom-mediated spin transfer effects. We also show that the demagnetization and spin transfer in the ferromagnetic–nonmagnetic CrSBr-P vdW heterostructure can be sensitively manipulated by laser pulses with different fluences. Our study offers a microscopic understanding of spin dynamics in these vdW heterostructures aiming toward their potential spintronic applications, which rely on optically controlled spin transfer processes.



Advances in spintronics are expected to significantly boost modern industry by enabling more efficient computing systems.^{1–4} An efficient manipulation of spins in magnetic materials by external stimuli, such as magnetic fields, gate voltages, and laser pulses, is central for quantum computing applications.^{5–9} The ultrashort laser pulses are particularly effective at controlling magnetic order, demagnetization, and spin injection in bulk materials (such as metals, compounds, and alloys) on the femtosecond (fs) scale.^{10–12} Moreover, laser pulses can direct electron oscillations and transport in an electromagnetic field within extremely short time frames while requiring minimal energy input.¹³ These dynamics occur on the Angstrom and attosecond scales, which are understood to represent purely electronic dynamics that take place prior to nuclear motion.^{14–16} Additionally, the advent of X-ray free electron lasers (XFELs) has enabled time-resolved measurements of ultrafast processes using ultrashort X-ray pulses of less than 100 fs. These instruments are widely applied in fields like magnetization dynamics.^{17–19} Therefore, controlling spin with laser pulses holds significant potential for advancing high-speed computing and quantum information technologies.

Two-dimensional (2D) magnetic materials and related van der Waals (vdW) heterostructures have emerged as versatile platforms for studying ultrafast spintronics.^{20–26} Among various 2D materials, CrSBr stands out as an air-stable magnetic semiconductor, possessing antiferromagnetic (AFM) properties in the bulk phase with a high Néel temperature of 132 K. However, it exhibits ferromagnetism (FM) in the monolayer limit.^{27,28} The cleavage energy of

CrSBr is only 0.18 J/m², which is lower than that of CrI₃ (0.3 J/m²), indicating that the CrSBr monolayer could be easily exfoliated from its bulk phase.²⁹ CrSBr semiconductors have shown great potential for applications in spintronics, magnonics, and quantum information science.^{30–33} Specifically, the ferromagnetic stability and large hole mobility in CrSBr monolayer can be further enhanced by interfacing with a single-layer nonmagnetic (NM) phosphorene.³⁴ This combination potentially addresses the limitations of its practical use in next-generation semiconductor processing techniques. However, despite these attractive properties, the underlying physical mechanisms of laser-pulse-induced spin dynamics in CrSBr-related semiconductor heterostructures require further exploration.

In this work, we investigated the optically induced ultrafast spin transfer in the FM anisotropic material CrSBr and the CrSBr-phosphorene (CrSBr-P) vdW heterostructure subjected to ultrashort laser pulses using real-time time-dependent density functional theory (rt-TDDFT). Our computational results show that an ultrafast laser can induce demagnetization and optically induced spin transfer (OISTR) in Cr atoms

Received: November 13, 2024

Revised: December 30, 2024

Accepted: January 7, 2025

Published: January 13, 2025



within 20 fs. These dynamics are accompanied by significant spin transfer to the S and Br atoms, followed by a spin injection into the P layer through the interfacial atom-mediated spin transfer (IAMSTR) process. Our subsequent analysis elucidates the microscopic mechanisms driving spin dynamics in FM-NM semiconductor heterostructures and offers valuable insights into interatomic spin transfer behavior.

The initial optimized structures of single-layer CrSBr and phosphorene obtained with PBE density functional (see the Supporting Information) have resulting in lattice parameters of $3.53 \text{ \AA} \times 4.73 \text{ \AA}$ for CrSBr monolayer and $3.32 \text{ \AA} \times 4.57 \text{ \AA}$ for phosphorene, respectively, in agreement with previous works.^{35,36} The atomic and band structures of these stand-alone components are given in Figure S1. Next, four distinct model stacking configurations of CrSBr-P vdW heterostructures are considered, as illustrated in Figure S2. The stable structure, shown in Figure 1a,b, has lattice parameters of 3.53

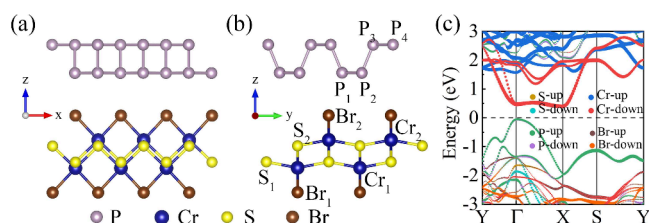


Figure 1. Panels a and b depict the atomic structure of the CrSBr-P vdW heterostructure viewed from different directions. (c) The projected band structure computed at the HSE06 level is decomposed into with spin-up and spin-down components. The contributing Cr-3d, S-2p, Br-2p, and P-2p orbitals are denoted with different colors.

$\text{\AA} \times 4.73 \text{ \AA}$ and an average lattice mismatch ratio of 4.66%. To evaluate the stability of this heterostructure, the binding energy is evaluated using the formula $E_b = E_{\text{CrSBr-P}} - E_{\text{CrSBr}} - E_{\text{P}}$. Here, $E_{\text{CrSBr-P}}$, E_{CrSBr} , and E_{P} represent the total energy of the

heterostructure, freestanding CrSBr monolayer, and phosphorene, respectively. The binding energy of the most stable model structure is calculated to be -0.13 eV per unit cell, indicating that CrSBr monolayer and phosphorene can spontaneously form a heterostructure without the external intervention. The interlayer distance between the CrSBr and phosphorene is $3.18 \text{ \AA}$. Furthermore, the initial magnetic moments of Cr, S, and Br atoms in the CrSBr-P vdW heterostructure are $2.76 \mu_B$, $-0.07 \mu_B$, and $0.03 \mu_B$, respectively. Here, μ_B stands for the Bohr magneton.

Figure 1c depicts the projected band structure of the CrSBr-P vdW heterostructure. The bands of NM phosphorene are split as a consequence of the magnetic proximity effects (MPE) between CrSBr and phosphorene.³⁷ Here, the HSE06 hybrid functional is employed to accurately evaluate the band gap of CrSBr-P vdW heterostructure, yielding a value of 0.45 eV . The valence band maximum (VBM) and the conduction band minimum (CBM) mainly originate from the phosphorene and Cr spin-down states. The global band alignment of the CrSBr-P vdW heterostructure facilitates efficient spin transfer from P to Cr. The electrons in the Cr-3d orbitals are distributed across an energy range of $>0.45 \text{ eV}$. Upon excitation by a femtosecond laser, electrons can first occupy the conduction bands of Cr-down orbitals due to the availability of empty states.³⁸

We next focus on the spin transfer dynamics of the CrSBr-P vdW heterostructure when it is excited by laser pulses. The femtosecond laser pulse most commonly used in experiments has an excitation energy of 2.3 eV (corresponding to a wavelength of 532 nm),^{39,40} a full width at half-maximum (fwhm) of 6.05 fs , and a fluence of 28.2 mJ/cm^2 , as shown in the gray line in Figure 2a. Figure S3 illustrates the demagnetization of Cr, S, and Br atoms in the CrSBr monolayer and CrSBr-P vdW heterostructure. Here the demagnetization of Cr atom in the heterostructure is more pronounced and occurs more rapidly than that in the

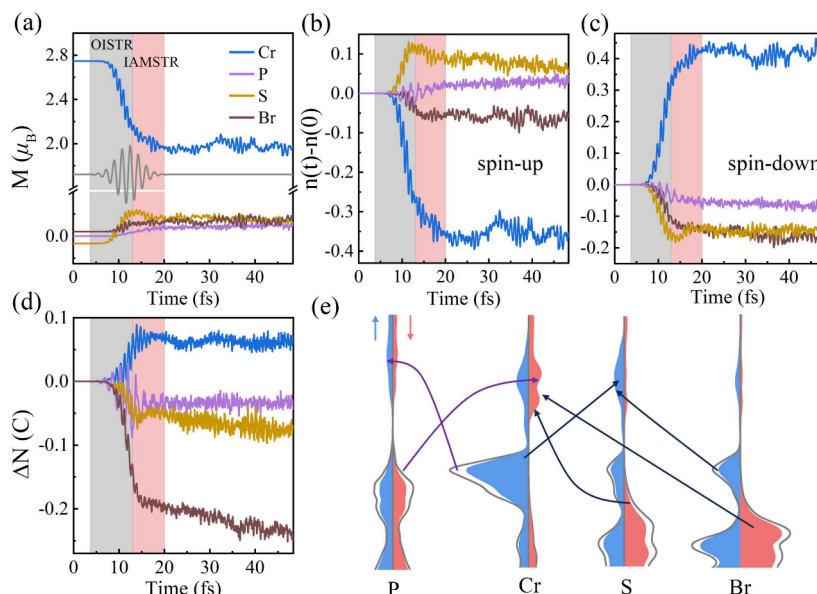


Figure 2. (a) Time-dependent dynamics of local magnetism for each element. The gray line represents the vector potential $A(t)$ of the laser pulse. (b and c) Change of spin-up and spin-down electrons ($n(t) - n(0)$) as a function of time. (d) Time-dependent charge dynamics of CrSBr-P vdW heterostructure. (e) Schematic diagram of the electron population during the photoexcitation process. The gray line, blue and red areas represent the initial state and the spin-up, spin-down states after excitation. The purple and black arrows show the transport of interlayer and intralayer spin electrons, respectively.

monolayer due to the enhanced spin-polarized charge transfer from the Cr-3d orbitals to the P-2p orbitals. We next inspect the demagnetization of each atom in the heterostructure when the atom is excited by a laser with an A_x or A_y polarization vector, as illustrated in Figure S4. The findings reveal slight discrepancies in the charge transfer and demagnetization of each atom in the two directions. Consequently, we will focus the subsequent discussion on the A_y polarization vector.

Given the pivotal role of the spin-orbit coupling (SOC) effect in the demagnetization process,⁴¹ the evolution of the local moment dynamics of the Cr atom and the time-dependent occupation changes for each atom in the CrSBr-P vdW heterostructure are calculated (Figure S5). The results reveal the emergence of the out-of-plane FM order in CrSBr layer, in line with previous reports.^{42,43} Upon the activation of the SOC, the concurrent occurrence of spin-transfer and spin-flip processes is enabled,⁴⁴ resulting in the observation of two primary modes of electron motion: (1) Before 15 fs, both spin-transfer and spin-flip processes are active, involving the transfer of electrons both within and between the layers. The transfer of electrons between Cr and S occurs first, followed by the sequential transfer to Br and P atoms. (2) The demagnetization process is predominantly driven by spin-flip. The magnetic moments of the S, Br, and P atoms remain essentially stable, with only flipping of the Cr-up to Cr-down electrons taking place. Moreover, the amplitude of demagnetization of Cr is slightly larger than that observed when SOC is not taken into account. The stable magnetic moments of S, Br, and P are nearly identical to those observed in the absence of SOC. Furthermore, as there are no phonon-mediated spin-flip processes, our simulations exclusively consider pure spin electron dynamics within a 50 fs time scale.

Upon excitation, the magnetic moments of Cr₁ and Cr₂ atoms decrease by approximately 0.80 and 0.85 μ_B , respectively, while each S, Br, and P atom gains roughly 0.20, 0.10, and 0.10 μ_B , respectively (see Figure S6a). The reduction in magnetic moments of the Cr atom is more pronounced than the increase observed in the magnetic moments of the other atoms. This discrepancy indicates the presence of excited interstitial electrons, which represent the intermediate state of electrons during transfer, where no interaction occurs with any moving nucleus.⁴³

After laser pulse irradiation, the atomic magnetic moment takes longer to reach equilibrium than the pulse's fwhm (Figure 2a). Moreover, $A(t)$ of the laser pulse begins to change around 4 fs, while the magnetic moment responds with a delay, starting at 7 fs, exhibiting a lag in its response to $A(t)$. To summarize, the demagnetization and spin injection processes among atoms can be divided into the following stages: (1) The local magnetism of Cr is reduced around 7 fs. (2) Cr undergoes demagnetization while the S magnetic moment increases. Moreover, the injection of spin into Br and P is delayed by ~ 1 fs and occurs in a sequential manner. (3) The increase in the magnetic moment of S and Br reaches a maximum at about 13 fs and then stabilizes. During this interval, the OISTR mechanism is primarily responsible for facilitating the transfer of magnetic moments from Cr atoms to S and Br atoms. (4) Cr and P atoms reach equilibrium at approximately 20 fs. The spin injection process between Cr and P is predominantly governed by the IAMSTR mechanism. Laser-induced spin dynamics indicates that the process of interlayer spin injection occurs at a slower rate than that of intralayer spin transfer. Notably, the calculated demagnet-

ization per Cr atom exceeds the experimentally observed value in the CrI₃/WSe₂ heterostructure, and the demagnetization occurs at a shorter time scale.⁴³

To gain insight into the microscopic mechanism of spin dynamics, it is critical to analyze the change in spin-polarized occupation of charge, as illustrated in Figure 2b,c. The occupations for spin-up and spin-down charges are defined as follows:

$$n(t) - n(0)\{\uparrow\} = \frac{1}{2}[M(t) + N(t)]$$

$$n(t) - n(0)\{\downarrow\} = \frac{1}{2}[N(t) - M(t)]$$

where $n(t) - n(0)$ is the time-dependent change of the local charge; M and N are the time-dependent dynamics of the local magnetism and charge, respectively.

The obtained time-dependent charge dynamics is shown in Figure 2d. The demagnetization process of Cr relies on the loss of spin-up electrons and the concomitant gain of spin-down electrons. In contrast to Cr atoms, the reduction in spin-down electrons for the other three atoms exceeds the increase in spin-up electrons, resulting in an overall increase in their magnetic moments. Furthermore, as demonstrated in Figure 2d, the valence electrons of the S, Br, and P atoms exhibit a population decrease following excitation, whereas those of the Cr atoms tend to gain. The magnetic moment is transferred from Cr to S, Br, and P, while valence electron transfer occurs in the opposite direction. The principal transport mechanism of spintronics is schematically illustrated in Figure 2e, including the following stages: the demagnetization of Cr is attributed to the transfer of the electrons from S-, Br-, and P-down to Cr-down, as well as Cr-up to P-up and S-up. The alteration in the magnetic moment of S is primarily assigned to the transfer of electrons from S-down to Cr-down, along with the transfer of electrons from Cr-up and Br-up to S-up. The observed increase in the magnetic moment of Br and P can be attributed to the outflow of spin-down electrons. Therefore, the laser-induced spin dynamics process is primarily driven by the transfer of spin electrons.

The real-space spin density and charge accumulation (or depletion) around the nucleus can be monitored in real time, as shown in Figure 3. It is evident that upon the activation of the laser pulse, the electrons are getting detached from the nucleus and undergo a transition to higher-energy excited states. As shown in Figure 3a, the reversal of the magnetic order of S is observed to occur initially upon laser pulse irradiation. Concurrently, the spin densities of all atoms are oriented in the same direction. Next, the atomic magnetic moment is getting transferred from Cr to Br and finally to P. The occupancy of electrons around an atom at a given moment provides an intuitive representation of the spatial distribution and movement of electrons near the nuclei, as shown in Figure 3b. It can be observed that within the 7–15 fs time-interval the electrons exhibit oscillatory behavior in sync with $A(t)$, a phenomenon that is most evident when the pulse intensity reaches its peak. This is evidenced by the significant accumulation and depletion of electrons at 12.10 fs. Here, the electron oscillations around P and Br are particularly noteworthy. Following stabilization, P (Br) atoms undergo a depletion (accumulation) of spin electrons.

Figure 4 illustrates the projected density of states (PDOS) occupation function of each atom at a specific time.

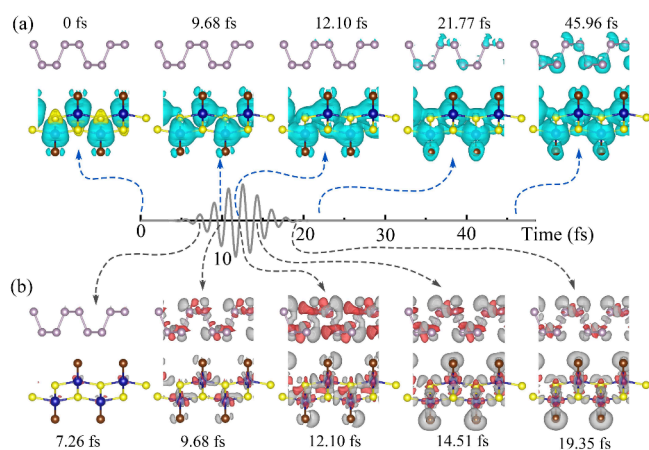


Figure 3. (a) Snapshots of the spin density of the CrSBr-P vdW heterostructure at different moments ($t = 0, 9.68, 12.10, 21.77$, and 45.96 fs), where cyan and yellow domains indicate electron spin-up and spin-down, respectively. (b) Snapshots of the charge-density difference at $t = 7.26, 9.68, 12.10, 14.51$, and 19.35 fs. Here regions of electron depletion and accumulation are gray and pink, respectively. The isosurfaces are depicted with a cutoff value of $0.003 \text{ e}/\text{\AA}^3$.

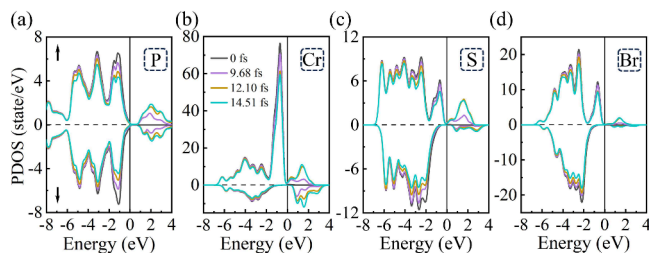


Figure 4. Time-resolved occupation function PDOS(t) of (a) P, (b) Cr, (c) S, and (d) Br channels are shown at selected $t = 0, 9.68, 12.10$, and 14.51 fs times.

Importantly, the number of unoccupied states determines the quantity of spin-polarized electrons that can be preserved during the initial stage of spin transfer. During the 9.68 – 14.51 fs time interval, electrons in the valence band of all elements undergo a partial transition to the conduction band, resulting in a gradual increase in the number of electrons occupying the conduction band. It has been observed that the spin-down PDOS in the Cr conduction band is larger than the respective spin-up PDOS. Consequently, the number of spin-down electrons stimulated to transition exceeds that of spin-up electrons, thus inducing the demagnetization phenomenon. In parallel, the magnetic moment of S increases. Furthermore, an examination of the static band structure (Figure 1c) reveals that the unoccupied band of Cr and the spin-up unoccupied band associated with S atoms can accept electrons from the occupied states of Cr, P spin-up, and Br spin-down.

We further examine the impact of laser parameters on the spin dynamics of the CrSBr-P vdW heterostructure. The three principal experimental parameters of ultrashort laser pulses are wavelength, amplitude, and a full width at fwhm, which directly determine the magnitude of the fluence. Figure 5 illustrates the magnetic moment and spin transfer at the Cr-atom in the context of laser-induced dynamics as a function of these parameters. The full range of laser pulse parameters used is provided in Table S1. A significant demagnetization of Cr is observed for shorter wavelengths (Figure 5a) and with larger amplitudes (Figure 5b). In contrast to the effects of wavelength

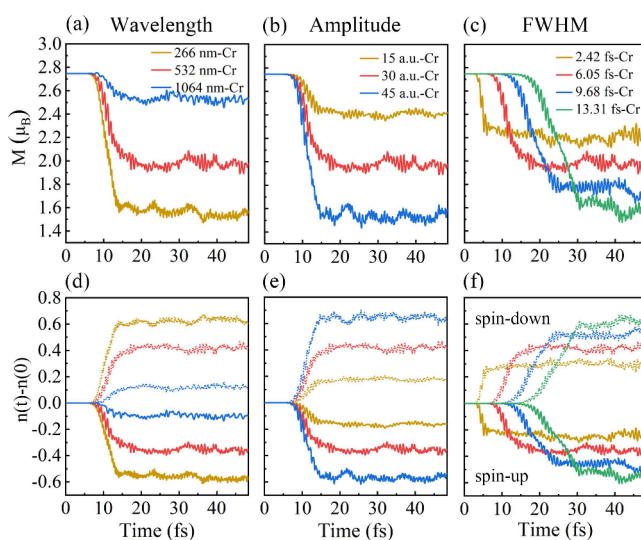


Figure 5. Control of laser-induced spin dynamics via wavelength, amplitude, and fwhm of laser: (a–c) time-dependent magnetic moment dynamics of Cr atom with respect to these 3 parameters, respectively, (d–f) variation of spin-up and spin-down electron populations ($n(t) - n(0)$) of Cr atom. The solid and dashed lines represent the spin-up and spin-down electrons.

and amplitude, the pulse width not only influences the degree of demagnetization but also affects the duration of the spin dynamics (Figure 5c). As the fwhm increases, the magnetic moment experiences a more pronounced change, illustrating how the laser pulse duration is capable of controlling spin dynamics on ultrashort time scales. The degree of demagnetization is typically contingent upon the fluence of the laser pulse. The impact of the laser parameters on the time-dependent spin electronic transformations of Cr, S, Br, and P atoms is illustrated in Figure 5d–f and Figure S7, respectively. Here, concurrent with the demagnetization of atoms, spin electron transfer is also observed. These findings suggest that all laser parameters could be effectively employed to control spin transfer in magnetic materials.⁴⁵

Variations of these parameters significantly affect the fluence of the laser pulse. For example, the fluence ($\sim 28 \text{ mJ}/\text{cm}^2$) and fwhm (6.05 fs) can be achieved by modifying the laser pulse wavelength and amplitude, thereby ensuring a specified energy density and action time. Table S1 summarizes the parameters of our simulations. The Cr demagnetization is not directly proportional to the wavelength or amplitude of the laser pulse, as shown in Figure S8. Instead, it is essential to optimize these two parameters to facilitate spin transfer in the material. Consequently, the Cr demagnetization process can be enhanced by reducing (increasing) the wavelength (amplitude) of the laser parameters.

Although the spin dynamics can be precisely controlled by parameters of the optical pulsed laser, it would be difficult to spatially resolve these dynamics optically. In this context, modern X-ray-free electron laser facilities (XFELs) have become a powerful tool for probing photoinduced dynamics with unprecedented resolution in space and time. For example, these instruments allowed detection of electronic charge motion occurring on a time scale of a few femtoseconds to attoseconds, i.e., attosecond charge migration processes.^{46–48} In this work, the rt-TDDFT calculations only focused on the pure electron dynamics by freezing the motion of the nucleus

adiabatically at a time scale of less than 50 fs for the CrSBr-P vdW heterostructure. This is based on the assumption that the lattice vibrations do not occur at this time scale within various molecular and 2D vdW systems, as reported in phenylalanine,¹⁴ MoS₂/MnSe₂,³⁸ and CrI₃/WSe₂⁴³ heterostructures. Nevertheless, beyond 100 fs, phonon-mediated spin dynamics processes are expected to produce a significant change in the magnetic moment of materials. The full nuclear dynamics calculations encompass the movement of atomic nuclei in response to the interaction of all phonon vibration modes upon laser irradiation. This approach was utilized to calculate the CrSBr-P heterostructure and further confirm the robustness of observed spin dynamics and magnetization on the time scale of less than 50 fs, as verified by the small spin moment difference of $\sim 0.3 \mu_B$ for Cr and negligible for S, Br, and P atoms (Figure S6b).

In addition, considering that changing the halogen element from Br to Cl and to I may produce significant variations in structural and electronic properties due to the differences in atomic size and electronegativity, we performed additional calculations (structure, electronic properties, and spin dynamics) on the CrSCl and CrSI monolayers. The results show a normal change with the increase of halogen atomic radius (Cl, Br and I), verified by a gradual increase in *a*, decrease in *b*, and decrease in band gap (shown in Table S2). Furthermore, the excitation effects on the three materials are analogous once the laser energy is beyond the band gap. It can be observed that the demagnetization time and amplitude for the CrI atom are essentially close to each other among the three CrSX (X = Cl, Br, and I) monolayers (Figure S10), arising from the similar band dispersion (Figure S9).

In conclusion, we performed *ab initio* simulations for optimal geometry to evaluate the ground-state properties and ultrafast photoinduced spin transfer dynamics during the first 50 fs in the NM-FM CrSBr-P vdW heterostructure. Our results indicate that the spin-polarized band structures of the NM-FM CrSBr-P vdW heterostructure and the proximity-induced magnetism observed in NM materials can be attributed to the influence of the FM CrSBr layer. The demagnetization of Cr atoms in CrSBr is less pronounced compared to that in the CrSBr-P vdW heterostructure. This effect can be rationalized by the transfer of a portion of the CrSBr spin to the phosphorene layer of the NM. Furthermore, the spin injection from the Cr atoms to the P atoms is completed within a few femtoseconds in the CrSBr-P vdW heterostructure, driven by a combination of the OISTR and IAMSTR effects. We further show that the demagnetization and the spin transfer in the FM-NM CrSBr-P vdW heterostructure can be sensitively manipulated by adjusting laser parameters, including the wavelength, amplitude, and pulse width. Overall, our results further elucidate the phenomena of proximity magnetism and spin transfer in CrSBr-P vdW heterostructures using optical tools and highlight their promise for potential applications in ultrafast spintronics.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpclett.4c03279>.

Computational details; atomic and band structure of CrSBr monolayer, phosphorene, and CrSBr-P vdW heterostructure; time-dependent dynamics of the local

magnetic moment and majority and minority electrons of different atoms and in CrSBr monolayer and CrSBr-P vdW heterostructure; time-dependent dynamics of magnetic moment subject to the full-phonon nuclear dynamics upon ultrafast laser irradiation; dynamics of magnetic moment and change in spin-resolved electrons as a function of the wavelengths, amplitudes, and fwhm; pulsed laser parameters utilized in the calculations of spin dynamics (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work is supported by National Natural Science Foundation of China (12374057, 12204069), the Startup funds of Outstanding Talents of UESTC (A1098531023601205), Sichuan Science and Technology

Program (2024NSFSC1385), and Fundamental Research Funds for the Central Universities (ZYGX2022J022). The work (S.T.) at Los Alamos National Laboratory (LANL) was performed in part at the Center for Integrated Nanotechnologies (CINT), a U.S. Department of Energy, Office of Science user facility at LANL.

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Supporting Information

Ultrafast Laser Induced Spin Dynamics in All-Semiconductor

Ferromagnetic CrSBr-Phosphorene van der Waals

Heterostructure

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Computational details

The ground state structural optimization and static electronic properties are calculated based on the density functional theory (DFT) integrated in the Vienna ab initio simulation package (VASP).¹ The exchange-correlation function was constructed by Perdew-Bruke-Ernzerhof (PBE) for the generalized gradient approximation (GGA).² The projected augmented wave method (PAW) was employed to describe the interaction between the atomic core and the valence electrons,³ while the vdW interaction was introduced by empirical dispersion correction opt88-vdW approach. The hybrid Heyd-Scuseria-Ernzerhof (HSE06) functional was used to calculate the

electronic structure for the CrSBr-P vdW heterostructures.⁴ A correction (GGA+U) to describe effect of Cr-3d electrons of $U = 3$ eV was employed.⁵ The kinetic-energy cutoff of 550 eV and the k-point mesh of $21 \times 17 \times 1$ was employed in the calculations. To avoid the artificial interactions between periodic images, a vacuum space of 30 Å was inserted along the z direction. The self-consistent atomic force convergence standard was 0.01 eV/Å and the energy convergence was 10^{-5} eV.

Rt-TDDFT were performed to reveal the spin dynamics of CrSBr and CrSBr-P vdW heterostructures based on it,⁶ and used to study the noninteracting partials in an effective potential by solving the time dependent Kohn-Sham (KS) equation. Specifically, the KS orbitals $\psi_j(\mathbf{r}, t)$ are Pauli spinors fulfilling the single-particle time-dependent Schrödinger equations as follows:

$$i \frac{\partial \psi_j(\mathbf{r}, t)}{\partial t} = \left[\frac{1}{2} \left(-i\nabla + \frac{1}{c} \mathbf{A}_{\text{ext}}(t) \right)^2 + v_s(\mathbf{r}, t) + \frac{1}{2c} \boldsymbol{\sigma} \cdot \mathbf{B}_s(\mathbf{r}, t) + \frac{1}{4c^2} \boldsymbol{\sigma} \cdot (\nabla v_s(\mathbf{r}, t) \times -i\nabla) \right] \psi_j(\mathbf{r}, t)$$

where $\mathbf{A}_{\text{ext}}(t)$ is the vector potential of the laser field in the dipole approximation. The effective potential is represented as $v_s(\mathbf{r}, t) = v_{\text{ext}}(\mathbf{r}, t) + v_H(\mathbf{r}, t) + v_{xc}(\mathbf{r}, t)$, which is consisted by external field $v_{\text{ext}}(\mathbf{r}, t)$, Hartree potential $v_H(\mathbf{r}, t)$ and the exchange-correlation (XC) potential $v_{xc}(\mathbf{r}, t)$, respectively. $\mathbf{B}_s(\mathbf{r}, t)$ is used to describe the KS magnetic field that is $\mathbf{B}_s(\mathbf{r}, t) = \mathbf{B}_{\text{ext}}(\mathbf{r}, t) + \mathbf{B}_{\text{XC}}(\mathbf{r}, t)$, $\mathbf{B}_{\text{ext}}(\mathbf{r}, t)$ and $\mathbf{B}_{\text{XC}}(\mathbf{r}, t)$ represent the magnetic field after applied laser pulse and an additional XC magnetic field, respectively. And the last term is stands for SOC.

The full non-collinear version of rt-TDDFT simulation is integrated in the full-potential augmented plane-wave ELK code were used to calculate the photoinduced dynamics. The laser pulse we used were linearly polarized (in-plane polarization) at a selected frequency. And all calculations were based on the adiabatic local spin density approximation (ALSDA).⁷ A regular mesh in k-space of $7 \times 5 \times 1$, a smearing width of 0.027 eV and a time step of 2.4 as were used for the calculation of spin dynamics.

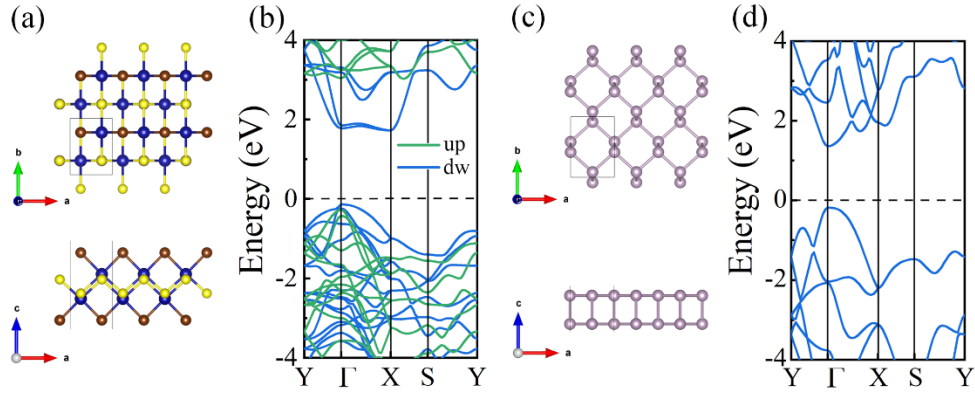


Figure S1. (a) Atomic and (b) band structure of CrSBr monolayer. (c) Atomic and (d) band structures of phosphorene, at the HSE06 level.

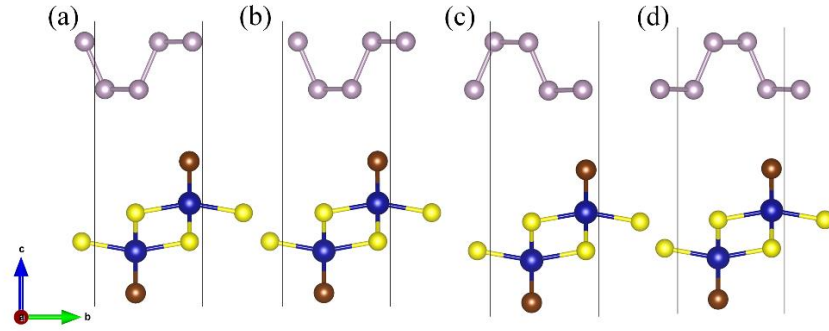


Figure S2. The atomic models of the CrSBr-P vdW heterostructure having a different dislocation of (a) 25, (b) 50, (c) 75 and (d) 100 % along b -axis direction, respectively.

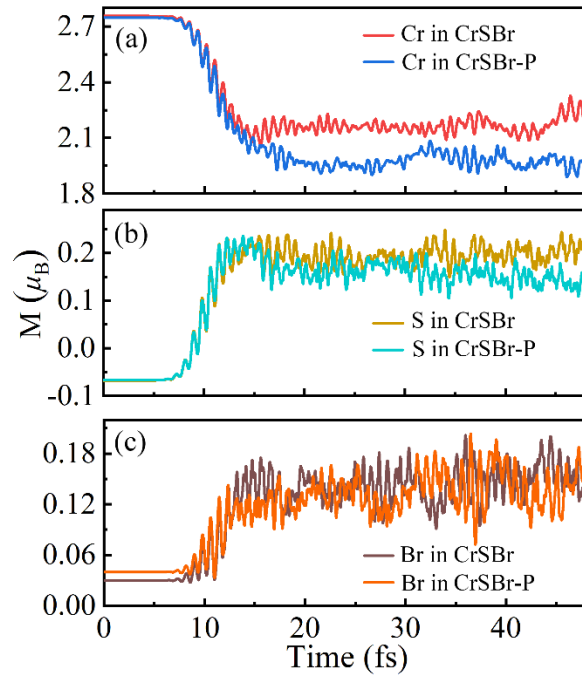


Figure S3. Time-dependent dynamics of the local magnetic moment for (a) Cr, (b) S, and (c) Br atoms in CrSBr monolayer and CrSBr-P vdW heterostructure, respectively.

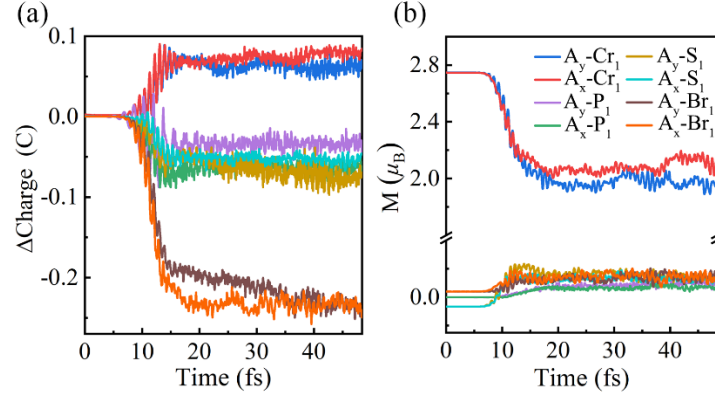


Figure S4. Time-dependent dynamics of (a) charge and (b) magnetic moment for each element along different polarization vectors (A_x and A_y) within CrSBr-P vdW heterostructure.

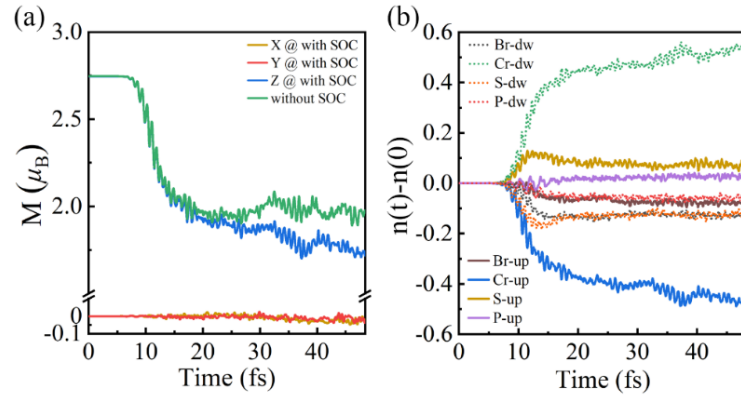


Figure S5. Time-dependent dynamics of (a) magnetic moment for Cr atom along A_y in CrSBr-P vdW heterostructure with and without SOC. (b) The evolution of the majority (up) and minority (down) electrons of Cr, S, Br and P atoms along y-direction in CrSBr-P vdW heterostructure with SOC.

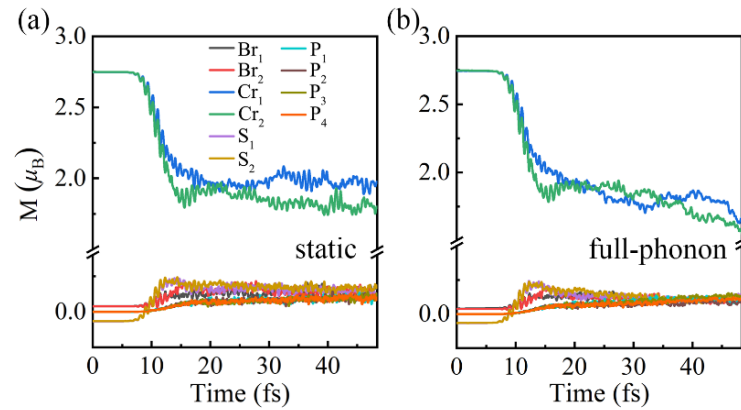


Figure S6. Time-dependent dynamics of magnetic moment for each atom along A_y in CrSBr-P heterostructure: (a) static nuclei frozen dynamics, (b) full-phonon nuclear dynamics upon ultrafast laser irradiation.

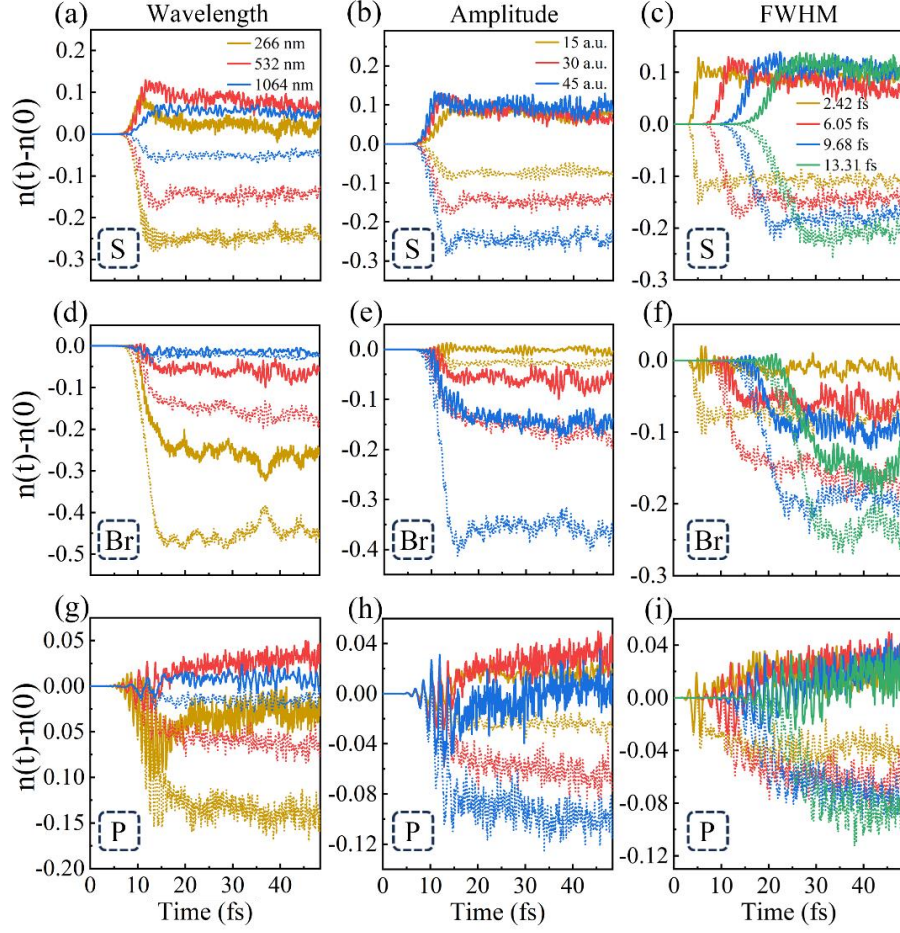


Figure S7. Time-dependent change in the spin-up and spin-down electrons of the (a-c) S, (d-f) Br, and (g-i) P atoms in CrSBr-P vdW heterostructure as a function of wavelengths, amplitudes, and full width at half maximum (FWHM). The solid line represents the spin-up electrons and the dashed line represents the spin-dw electrons.

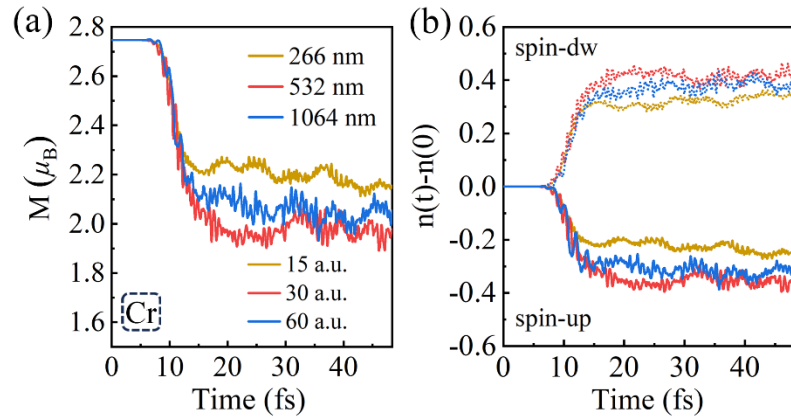


Figure S8. Time-dependent dynamics of the (a) magnetic moment and (b) change in spin-up and spin-dw electrons of the Cr atom in CrSBr-P vdW heterostructure with the same fluence and FWHM but with different wavelength and amplitude.

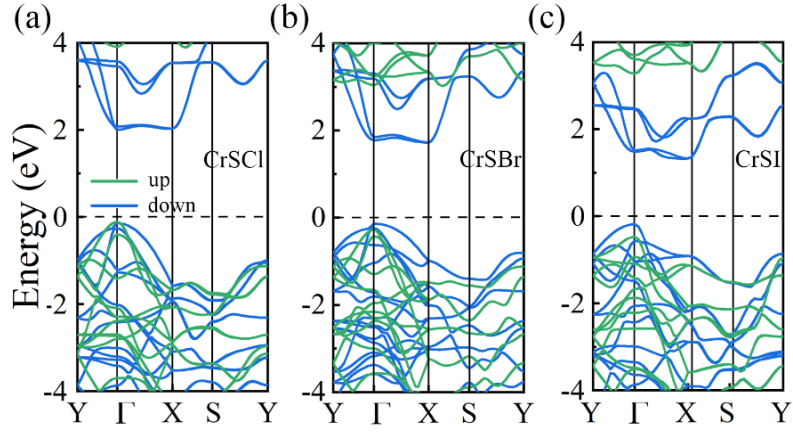


Figure S9. Band structures of (a) CrSCl, (b) CrSBr and (c) CrSI monolayer.

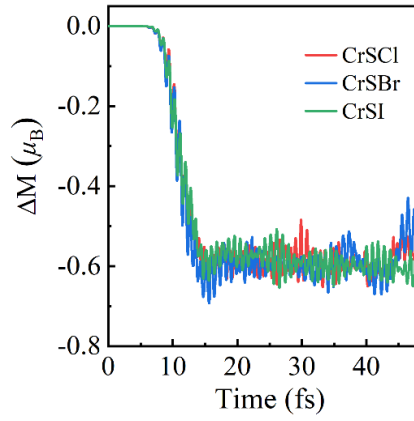


Figure S10. Time-dependent dynamics of magnetic moment change for Cr_1 along A_y in CrSX ($X = \text{Cl}, \text{Br}, \text{I}$).

Table S1. The pulsed laser parameters utilized in the calculations of spin dynamics of CrSBr and CrSBr-P vdW heterostructure.

Wavelength (nm)	Amplitude (a.u.)	FWHM (fs)	Fluence (mJ/cm^2)	Photon energy (eV)
266	30	6.05	112.40	4.66
532	30	6.05	28.21	2.33
1064	30	6.05	7.18	1.16
532	15	6.05	7.05	2.33
532	45	6.05	63.49	2.33
532	30	2.42	11.64	2.33
532	30	9.68	44.98	2.33

532	30	13.31	61.78	2.33
266	15	6.05	28.10	4.66
1064	60	6.05	28.73	1.16

Table S2 The lattice constants (a , b), Bader charge of Cr atom and band gap of CrSX (X = Cl, Br, I) monolayers.

	a (Å)	b (Å)	charge (e)	band gap (eV)
CrSCl	3.46	4.81	0.55	2.15
CrSBr	3.53	4.73	0.45	1.84
CrSI	3.73	4.79	0.36	1.51

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